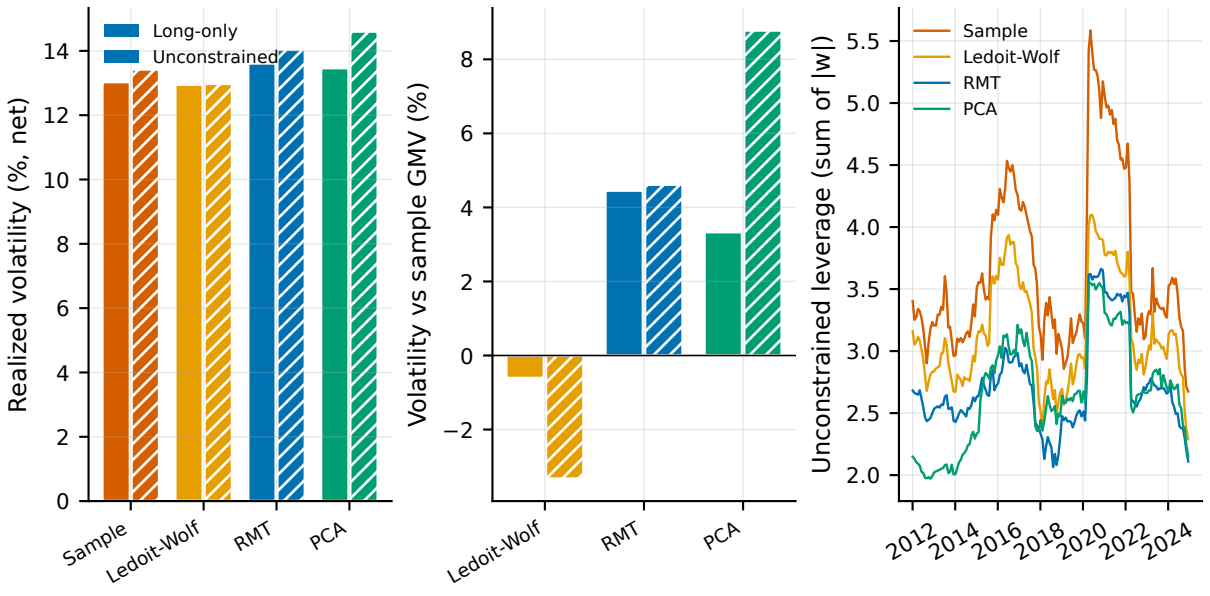
Long-only
Unconstrained
Realized volatility (%, net)
0
2
4
6
8
10
12
14
Sample
Ledoit-Wolf
RMT
PCA
Volatility vs sample GMV (%)
−2
0
2
4
6
8
Ledoit-Wolf
RMT
PCA
Unconstrained leverage (sum of |w|)
Sample
Ledoit-Wolf
RMT
PCA
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
2012
2014
2016
2018
2020
2022
2024